

DI - GAMES AND TOURNAMENTS

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

Fourteen teams which qualified for the Cricket World Cup are divided into two pools of seven teams each. In each pool, every team plays with every other team exactly once. Two points are awarded for a win and zero points for a loss. There are no draws in the tournament. At the end of all the matches, the top two teams in each pool, in terms of the points scored, advance to the semifinals. The winners of the semifinals play the finals and the winner of the final is crowned the World Champion.

Note: If two or more teams end up with the same number of points, the team with better net run rate is placed higher.

What is the total number of matches in the tournament?

- ☐ a) 43
- ☐ b) 45
- ☐ c) 46
- ☐ d) 48

Solution:

In each pool there are 21 matches. Then there are two semifinals and the final. The total number of matches is $21 + 21 + 2 + 1 = 45$ Choice (B)

What is the minimum number of points with which a team can advance to the semifinals?

- ☐ a) 4
- ☐ b) 8
- ☐ c) 6
- ☐ d) 2

Solution:

The minimum points with which a team can advance would happen when one of the teams in a pool wins all the matches and the other teams share the points nearly equally. As there are 21 matches in a pool, the total points available is 42. One team can win all its six matches and the remaining 15 matches can be shared by the six teams, three teams winning three each and the other three teams winning two each. So the team which has the best net run rate among the teams which won three matches, would advance to the semifinals. Choice (C)

What can be the maximum points scored by a team which was eliminated in the pool stage?

- ☐ a) 10
- ☐ b) 8
- ☐ c) 12
- ☐ d) 6

Solution:

The maximum points which a team can score and still be eliminated is when three teams end up with the maximum possible points and one of them is eliminated on the basis of net run rate. Three teams can win five matches each (10 points) and one of them would be eliminated.
Choice (A)

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

What is the maximum number of teams in a pool which can end up with six points each?

Solution:

As there are 21 matches in a pool, all the seven teams can win three matches each and have six points.
Ans: (7)

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

At most how many more matches can any team win than the winner of the tournament?

Solution:

A team can win just three matches in the pool stage and win the semifinals and finals and win the tournament, i.e. with just five wins. Another team can win all the matches in the pool stage and the semifinals but lose the finals, i.e., seven wins. So a team can win at most two matches more than the winner of the tournament.
Ans: (2)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

56 players are scheduled to take part in a tennis tournament, played on a knock out basis. The players are seeded from 1 to 56, with seed 1 being the top seed, seed 2 the next, and so on till seed 56, who is the last seeded player. In the first round, few of the top seeds are given a bye. There are no byes from the second round. The matches in any round are scheduled such that, if there are no upsets, the highest seeded player is scheduled to play the lowest seeded player left, the second seeded player is scheduled to play the second lowest seeded player left and so on.

An upset is said to happen if a lower seeded player beats a higher seeded player. In case of an upset, the matches take place as originally scheduled, except that the player who caused the upset takes the designated place of the player he upset.

What is the total number of matches in the tournament?

- ☐ a) 51
- ☐ b) 54
- ☐ c) 63
- ☐ d) None of these

Solution:

As there are 56 players and one player is eliminated in each match, there would be 55 matches in total as only one player is not eliminated.
Choice (D)

Which player could face the player seeded 3 in the third round?

- ☐ a) Seed 12
- ☐ b) Seed 14
- ☐ c) Seed 46
- ☐ d) More than one of the above

Solution:

The player seeded 3 would get a bye in the first round. In the second round, he would meet the player seeded 30 or the player seeded 35. In the third round, he would meet the player seeded 14 or 19 or 51 or 46.
Choice (D)

If a player who wins two sets in a match is the winner of the match, what is the minimum number of sets won by the winner of the tournament?

- ☐ a) 8
- ☐ b) 10
- ☐ c) 9
- ☐ d) 12

Solution:

The minimum number of matches to win the tournament is five, if any of the top eight seeds win the tournament. So the minimum number of sets to be won is 10.

Choice (B)

Which of the following seeded players could have met the player seeded 4 in the semifinals, assuming seed 4 reached the semifinals?

- ☐ a) 30
- ☐ b) 19
- ☐ c) 25
- ☐ d) 28

Solution:

The player scheduled to meet the player seeded 4 in the semifinals is seed 1. The other possibilities are all potential opponents of seed 1 from the previous rounds, i.e. seeds 32 or 33 or 16 or 17 or 49 or 48 or 8 or 9 or 25 or 24 or 40 or 41 or 56.

Choice (C)

If there are only three upsets in the tournament, who is the lowest seeded player who could have won the tournament?

- ☐ a) 7
- ☐ b) 8
- ☐ c) 15
- ☐ d) 16

Solution:

The lowest seeded player who can win with just three upsets would be seed 8 if he causes an upset in the quarter finals, semifinals and the finals. Choice (B)

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

Ram and Shyam are playing a game which involves picking up chocolates kept on a table. The players take turns alternately and each player in his turn has to pick up at least one and at most six chocolates in his turn. Assume that both players are equally intelligent and play to the best of their abilities to win the game.

Assume that the player who picks the last chocolate loses the game.

The game started with 75 chocolates on the table and it is Ram's turn to start. Which of the following can be the number of chocolates Ram should pick up so as to win the game, no matter how Shyam plays?

- ☐ a) 2
- ☐ b) 3
- ☐ c) 4
- ☐ d) 5

Solution:

The minimum number of chocolates to be picked up is one and the maximum number is six. The sum of these two numbers (7) is called the controlling factor. This is because the player who has the upper hand can ensure that in every round, seven chocolates are removed from the table. If one of the players can ensure that there are $7K + 1$ chocolates at the end of his turn, he can ensure that $7K$ chocolates are picked up after K rounds and the other player will have to pick up the last chocolate.

Ram has to ensure that the number of chocolates left on the table after his turn is of the form $7K + 1$. So he has to pick up 4 chocolates. Choice (C)

Assume that the player who picks the last chocolate loses the game.

During the game if Shyam faced a situation when he realized he cannot win the game no matter how many chocolates he picks up. Which of the following can be the number of chocolates on the table before his turn?

- ☐ a) 29
- ☐ b) 43
- ☐ c) 52
- ☐ d) More than one of the above

Solution:

The minimum number of chocolates to be picked up is one and the maximum number is six. The sum of these two numbers (7) is called the controlling factor. This is because the player who has the upper hand can ensure that in every round, seven chocolates are removed from the table. If one of the players can ensure that there are $7K + 1$ chocolates at the end of his turn, he can ensure that $7K$ chocolates are picked up after K rounds and the other player will have to pick up the last chocolate.

If the number of chocolates on the table before Shyam's turn is of the form $7K + 1$ he cannot win the game, no matter what he plays. Both 29 and 43 is of the form $7K + 1$.
Choice (D)

Assume that the player who picks the last chocolate loses the game.

During Shyam's turn he picked up four chocolates from the table which ensured that he won the game. Which of the following could have been the number of chocolates on the table before he picked up the four chocolates?

- ☐ a) 44
- ☐ b) 67
- ☐ c) 89
- ☐ d) None of these

Solution:

The minimum number of chocolates to be picked up is one and the maximum number is six. The sum of these two numbers (7) is called the controlling factor. This is because the player who has the upper hand can ensure that in every round, seven chocolates are removed from the table. If one of the players can ensure that there are $7K + 1$ chocolates at the end of his turn, he can ensure that $7K$ chocolates are picked up after K rounds and the other player will have to pick up the last chocolate.

If the number of chocolates on the table before Shyam's turn is of the form $7K + 5$, he needs to pick up four chocolates so that Ram is left with a number in the form $7K + 1$. Only 89 is of the form $7K + 5$.
Choice (C)

Assume that the player who picks the last chocolate wins the game.

During a game when it was Shyam's turn to play, there were 65 chocolates left on the table.

Which of the following is the number of chocolates that he should pick up so as to ensure his win?

- ☐ a) 2
- ☐ b) 1
- ☐ c) 3
- ☐ d) None of these

Solution:

The minimum number of chocolates to be picked up is one and the maximum number is six. The sum of these two numbers (7) is called the controlling factor. This is because the player who has the upper hand can ensure that in every round, seven chocolates are removed from the table. If one of the players can ensure that there are $7K + 1$ chocolates at the end of his turn, he can ensure that $7K$ chocolates are picked up after K rounds and the other player will have to pick up the last chocolate.

In cases where the person who picks up the last chocolate wins the game, a player can win if he ensures that the number of chocolates left on the table after his turn is of the form $7K$. He can then ensure that seven chocolates are removed from the table in each round and he can pick up the last chocolate. As the number of chocolates to be left on the table after his turn should be of the form $7K$, he should pick up two chocolates.

Choice (A)

Assume that the player who picks the last chocolate wins the game.

During his turn, Shyam picked up five chocolates which ensured his win. Which of the following could have been the number of chocolates on the table before he picked up the five chocolates?

- ☐ a) 65
- ☐ b) 32
- ☐ c) 40
- ☐ d) 67

Solution:

The minimum number of chocolates to be picked up is one and the maximum number is six. The sum of these two numbers (7) is called the controlling factor. This is because the player who has the upper hand can ensure that in every round, seven chocolates are removed from the table. If one of the players can ensure that there are $7K + 1$ chocolates at the end of his turn, he can ensure that $7K$ chocolates are picked up after K rounds and the other player will have to pick up the last chocolate.

The number of chocolates after his turn should be of the form $7K$. As he picked up five chocolates, the number of chocolates before he picked up was of the form $7K + 5$.

Choice (C)